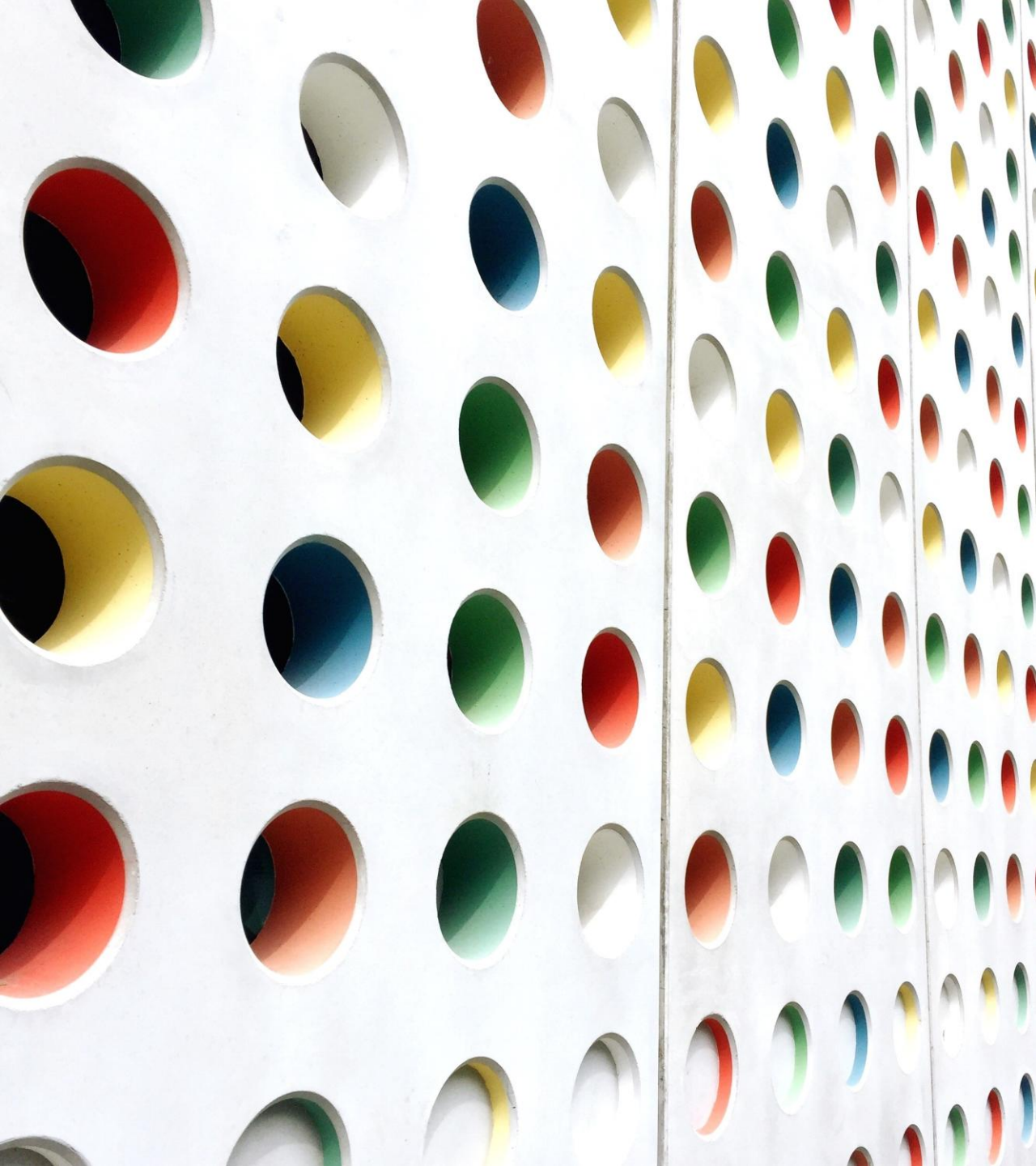




Utility Usage Prediction Tool

MACHINE LEARNING BASED
ELECTRICITY CONSUMPTION
PREDICTION

Introduction

- Predicts household electricity usage using Machine Learning.
- Uses electrical parameters like **Voltage** and **Global Intensity**.
- Helps estimate future electricity consumption.
- Built using **Python** and **Linear Regression**.

Problem Statement & Objectives

➤ Problem Statement

Manual estimation of electricity usage is difficult.

Inaccurate predictions may lead to poor energy planning.

Need for an automated prediction system.

➤ Objectives

Predict electricity usage accurately.

Automate the prediction process.

Apply Machine Learning techniques.

Improve energy consumption analysis.

Technology Stack

➤ Technologies Used

Python

Pandas

NumPy

Scikit-learn

Joblib

➤ Machine Learning Algorithm

Linear Regression

System Architecture

User Input



Data Preprocessing



Linear Regression Model



Prediction



Output

Machine Learning Workflow

Load Dataset



Data Cleaning



Feature Selection



Train-Test Split



Train Model



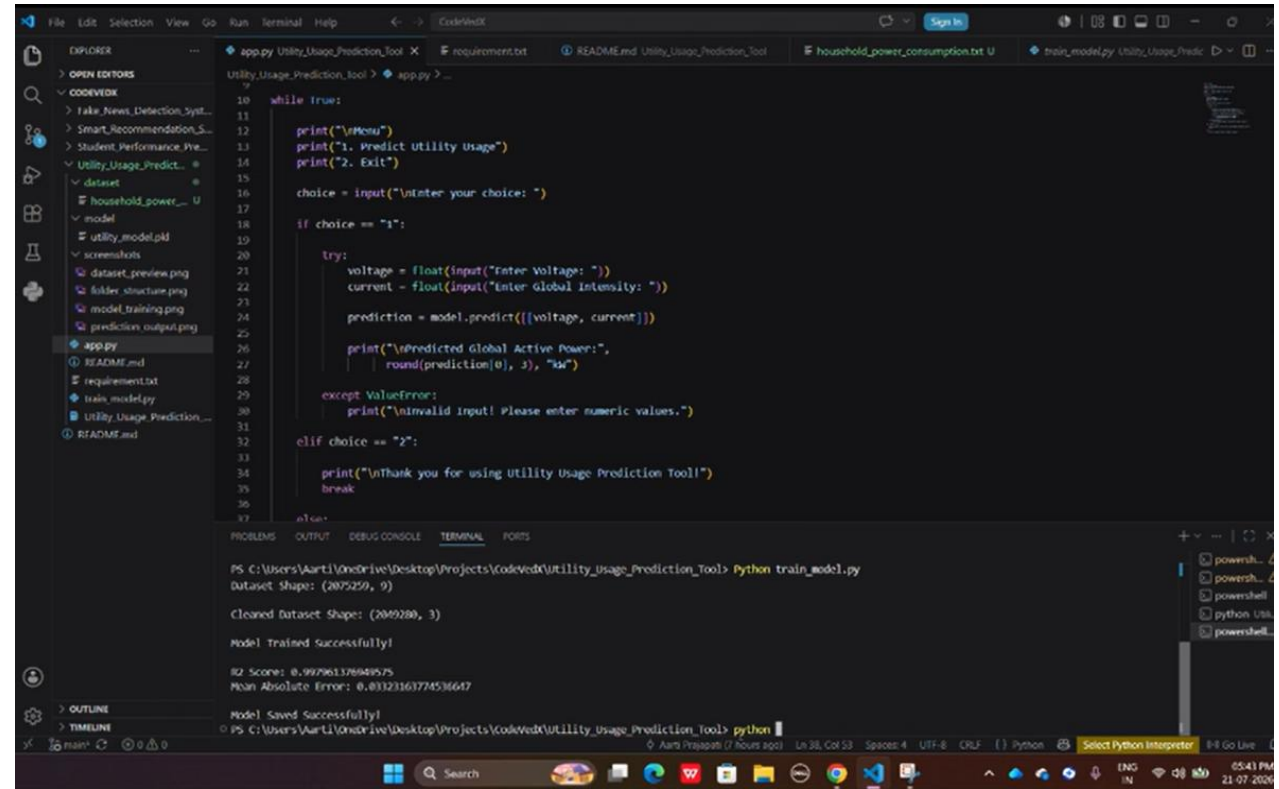
Save Model



Prediction

Project Features

- Console-based Application
- Data Preprocessing
- Machine Learning Prediction
- CSV File Handling
- Exception Handling
- Model Saving using Joblib



```
File Edit Selection View Go Run Terminal Help
Utility_Usage_Prediction_Tool X requirement.txt README.md Utility_Usage_Prediction_Tool household_power_consumption.txt U train_model.py Utility_Usage_Predic

EXPLORER
OPEN EDITORS
CODE EDITOR
Utility_Usage_Prediction_Tool X app.py > ...
10 while True:
11
12     print("\nMenu")
13     print("1. Predict Utility Usage")
14     print("2. Exit")
15
16     choice = input("\nEnter your choice: ")
17
18     if choice == "1":
19
20         try:
21             voltage = float(input("Enter Voltage: "))
22             current = float(input("Enter Global Intensity: "))
23
24             prediction = model.predict([[voltage, current]])
25
26             print("\nPredicted Global Active Power:",
27                   round(prediction[0], 3), "kW")
28
29         except ValueError:
30             print("\nInvalid input! Please enter numeric values.")
31
32     elif choice == "2":
33
34         print("\nThank you for using Utility Usage Prediction Tool!")
35         break
36
37
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Aarti\OneDrive\Desktop\Projects\CodeVed\Utility_Usage_Prediction_Tool> python train_model.py
Dataset Shape: (2075259, 9)
Cleaned Dataset Shape: (2069280, 3)
Model Trained Successfully!
R2 Score: 0.997961378949575
Mean Absolute Error: 0.03123161774536647
Model Saved Successfully!
PS C:\Users\Aarti\OneDrive\Desktop\Projects\CodeVed\Utility_Usage_Prediction_Tool> python
```

Results

- Successfully predicts electricity usage.
- Fast and efficient prediction.
- User-friendly console application.
- Demonstrates practical use of Machine Learning.

Conclusion

- Developed a Machine Learning-based Utility Usage Prediction Tool.
- Implemented **Linear Regression** for prediction.
- Improved understanding of data preprocessing and model training.
- Demonstrated a practical AI/ML application for energy usage prediction.

Thank You!